Guide to Interpreting the Output of the Fourier Analysis Code (Version 8)

1 Purpose of the code

The v. 8 code takes as input a list of pairs (n, μ_n), where μ_n is the minimum modulus among the eigenvalues of a certain matrix associated to the index n. Its purpose is to determine whether the sequence $\{\mu_n\}$ oscillates in a nearly periodic fashion and, if so, to estimate the period.

The code produces a text report (described in Sections 4–10) and six diagnostic plots (described in Section 11).

2 Glossary of abbreviations and acronyms

The following abbreviations appear in the code and its output.

ACF **Autocorrelation function.** Given a sequence $x_1, \ldots, x_N$ with mean $\bar{x}$, the (normalized) autocorrelation at lag k is

$$\mathrm{ACF}(k) = \frac{\sum_{i=1}^{N-k}(x_i - \bar{x})(x_{i+k} - \bar{x})}{\sum_{i=1}^{N}(x_i - \bar{x})^2}.$$

By construction $\mathrm{ACF}(0) = 1$. If the sequence is periodic with period T, then $\mathrm{ACF}(T)$ will be a local maximum close to 1. If the sequence is monotone or trend-like, the ACF decays from 1 toward 0 without oscillating.

AR(1) **First-order autoregressive model.** A simple stochastic model in which each value depends linearly on its predecessor plus noise:

$$x_n = \rho\, x_{n-1} + \varepsilon_n, \qquad \varepsilon_n \sim \mathcal{N}(0, \sigma^2).$$

The parameter ρ (reported in the output as "lag-1 autocorrelation" or simply "rho") measures how strongly consecutive values are correlated. The AR(1) model serves as a *null hypothesis*: it represents data that has short-range correlation but no true periodicity. A spectral peak is declared significant only if it exceeds what the AR(1) model would produce.

Bonferroni correction

A standard multiple-testing correction. When testing m frequency bins simultaneously, the probability of at least one false positive at level α can be as high as $m\alpha$. The Bonferroni correction replaces α with α/m at each bin, so that the *family-wise* false positive rate stays at α. In the code, m equals the number of frequency bins with period $\leq N/3$.

dB **Decibel.** A logarithmic unit used in the spectrogram (Section 11). Power in decibels is $10\log_{10}(\text{power})$.

FFT **Fast Fourier Transform.** An algorithm that computes the discrete Fourier transform of a length-N sequence in $O(N \log N)$ operations. The discrete Fourier transform decomposes the sequence into a sum of sinusoidal components at frequencies k/N for $k = 0, 1, \ldots, N-1$.

MA(w) **Moving average with window width w.** The smoothed value at index i is the arithmetic mean of the w values centered at i. Used in the multi-scale period detection (Section 8) to suppress fine-scale structure before re-measuring the ACF.

SG **Savitzky–Golay filter.** A smoothing method that fits a low-degree polynomial (here, cubic) by least squares within a sliding window of specified width, and replaces each data value with the fitted value at that point. Unlike a moving average, it preserves local peaks and inflection points better.

N The number of data points in `mins_list`.

$N/3$ The maximum period considered credible. A period of T observed in N data points corresponds to N/T full cycles; for $T > N/3$ there are fewer than three full cycles, which is generally insufficient to distinguish true periodicity from a trend artifact.

rho (ρ) The lag-1 autocorrelation of the detrended data, used as the parameter of the AR(1) null model. Values near $+1$ indicate strong positive serial correlation (each value close to its predecessor); values near -1 indicate alternation; values near 0 indicate near-independence.

std dev Standard deviation of the detrended residuals.

3 Section 1 of the report: data summary

The report begins by printing the range of the index n, the number of data points N, and the range of the minimum moduli μ_n. These are for basic sanity-checking.

4 Section 2 of the report: adaptive detrending

Before searching for periodicity, the code removes any broad non-oscillatory trend from the data, because a monotone growth curve (for instance) would inject spurious low-frequency power into the Fourier transform.

Stage A: polynomial detrend

A degree-2 polynomial (parabola) is fitted to the data by least squares and subtracted. This is gentle enough not to absorb any oscillation whose period is less than $N/3$.

The code then examines the residuals to decide whether Stage A was sufficient. It computes the ACF of the residuals and checks two things:

1. Does the ACF cross zero within the first $N/3$ lags? If so, the residual is oscillatory, which is the desired outcome: the polynomial removed drift without destroying periodicity.

2. Is the lag-1 autocorrelation above 0.85? If the ACF does not cross zero and the lag-1 autocorrelation exceeds 0.85, the residual is still dominated by a trend that the parabola could not capture.

Stage B: Savitzky–Golay filter (conditional)

Stage B is invoked only when Stage A leaves a trend-like residual (no zero crossing *and* lag-1 autocorrelation exceeding 0.85). This occurs, for example, when the data follows a ramp-then-plateau shape that no single polynomial fits well.

The code applies a Savitzky–Golay filter with a cubic polynomial and tries several window widths, selecting the one that minimizes the lag-1 autocorrelation of the resulting residual. Crucially, the window width is capped at $N/3$ to prevent the smoother from absorbing a genuine oscillation whose period is shorter than the window.

The report prints which detrending method was selected and the standard deviation of the final detrended residuals.

5 Section 3 of the report: windowed FFT

The detrended data is multiplied by a *Hann window* (a bell-shaped taper that goes to zero at both ends) before computing the FFT. The purpose of windowing is to reduce *spectral leakage*: without it, the FFT implicitly assumes the data is periodic over its full length, and any mismatch at the endpoints spreads spurious power across all frequencies. A multiplicative correction factor compensates for the power lost to the window.

The output of the FFT is a *power spectrum*: for each frequency $f = k/N$, a non-negative number measuring how much of the data's variance is attributable to oscillation at that frequency. The corresponding period is $T = 1/f$.

6 Section 4 of the report: significance testing

Not every bump in the power spectrum is meaningful; random noise also produces spectral peaks. To distinguish real periodicity from noise, the code constructs a null model and asks: could the observed peak have arisen by chance?

The AR(1) null model

The null hypothesis is that the detrended data is generated by a first-order autoregressive process (see the glossary). The theoretical power spectrum of an AR(1) process with lag-1 autocorrelation ρ and variance σ^2 is

$$S(f) = \frac{\sigma^2(1-\rho^2)}{1 - 2\rho\cos(2\pi f) + \rho^2},$$

following Gilman, Fuglister, and Mitchell (1963). This spectrum is not flat: when $\rho > 0$ it is tilted toward low frequencies ("red noise"), and when $\rho < 0$ it is tilted toward high frequencies. The normalization ensures that the mean of the theoretical spectrum matches the mean of the observed spectrum.

Chi-squared test with Bonferroni correction

Under the null model, the power at each frequency bin is approximately distributed as a scaled chi-squared random variable with 2 degrees of freedom. The 95% and 99% confidence thresholds are computed from this distribution, then inflated by the Bonferroni correction to account for the number of bins tested (see the glossary).

The report prints the number of frequency bins tested and the Bonferroni-corrected per-bin significance levels.

7 Section 5 of the report: significant FFT peaks

The code identifies all local maxima of the power spectrum that exceed the Bonferroni-corrected 95% or 99% confidence threshold, restricted to periods $\leq N/3$. These are listed in a table with columns:

Rank Ordered by power (strongest first).

Period $T = 1/f$, measured in index units (i.e., if Period = 60, the pattern repeats every 60 values of n).

Frequency $f = 1/T$.

Power The squared magnitude of the Fourier coefficient at this frequency, displayed in scientific notation (e.g., `1.58e+04` means $1.58 \times 10^4 = 15{,}800$).

Signif. Whether the peak exceeds the 95% or 99% threshold.

Important caveat. For non-sinusoidal waveforms (sawtooth, square-wave, or similar shapes built from sharp segments), the FFT distributes the oscillation's energy across many *harmonics* (integer multiples of the fundamental frequency). The highest-power FFT peak may therefore correspond to a harmonic rather than the fundamental period. The multi-scale analysis described next addresses this issue.

8 Section 5b of the report: multi-scale period detection

This section identifies the *fundamental period* by a method that is immune to the harmonic ambiguity of the FFT.

The idea is to examine the data at progressively coarser scales by smoothing it with moving averages of increasing width $w = 1$ (raw), $3, 5, 7, \ldots$ At each scale, the ACF of the smoothed signal is computed, and the first ACF peak above lag w with strength exceeding 0.1 is recorded. (The restriction to lags above w prevents the smoothing itself from creating artificial peaks.)

At the finest scale (raw data), the ACF detects the shortest repeating feature, which may be sub-cycle structure (e.g., the individual teeth of a sawtooth). At coarser scales, the fine structure is averaged out and only the true repeat period survives.

The results are collected into a table with columns:

Period The lag at which the ACF peaks, in index units.

Scale Which smoothing was applied. "raw" means no smoothing; "MA(w)" means a moving average of width w.

ACF strength
The value of the normalized ACF at the detected lag. Ranges from -1 to $+1$; values close to $+1$ indicate a strong, clean repetition. Values below 0.2 are considered too weak to be reliable.

Role The longest detected period is labeled FUNDAMENTAL; shorter periods are labeled "sub-cycle structure."

9 Harmonic analysis table

When both a fundamental period and significant FFT peaks are present, the code checks whether each FFT peak is a *harmonic* of the fundamental. The table shows:

$T_{\text{fund}}/T_{\text{peak}}$ The ratio of the fundamental period to the FFT peak's period. If this is close to a positive integer m, the FFT peak corresponds to the m-th harmonic.

Nearest int The integer nearest to the ratio.

Harmonic? If the ratio is within 15% of an integer, the peak is tagged as "$\sim m$x" (an approximate m-th harmonic). Otherwise it is tagged "independent," meaning it represents a genuinely distinct period not explained as a harmonic of the fundamental.

A waveform whose FFT peaks are all harmonics of the fundamental is non-sinusoidal but still truly periodic with the fundamental period.

10 Summary

The final section of the report draws one of the following conclusions:

1. **Fundamental period reported, with ACF strength above 0.2.** This is the main positive result: the data oscillates with the stated period. Any sub-cycle periods are also listed.

2. **Weak periodicity (ACF strength ≤ 0.2).** A period was detected at some scale, but the autocorrelation is too weak to be confident.

3. **No statistically significant periodicity.** Neither the FFT (after Bonferroni correction) nor the multi-scale ACF found evidence of repeating behavior.

11 The six diagnostic plots

1. **Original: Minimum Moduli.** The raw input data (n, μ_n). Visual inspection of this plot is the most basic sanity check.

2. **Data with Trend.** The raw data (blue) overlaid with the fitted trend (red). Check that the red curve captures the broad shape without tracking the oscillations.

3. **Detrended Residuals.** The data after subtracting the trend. If detrending worked well, this should oscillate around zero with no visible upward or downward drift.

4. **Power Spectrum with Bonferroni-Corrected Significance.** A log-scale plot of power versus frequency. The dashed curves are:

   - black: the AR(1) red-noise spectrum (the null model),
   - green: the 95% Bonferroni-corrected threshold,
   - red: the 99% Bonferroni-corrected threshold.

   Peaks marked with colored dots exceed the corresponding threshold. This plot is the primary evidence for or against periodicity.

5. **Power Spectrum vs. Period.** The same information as plot 4, but with the horizontal axis showing period rather than frequency. This is often easier to read because periods are directly interpretable ("the pattern repeats every T steps").

6. **Spectrogram.** A time–frequency decomposition showing how the spectral content changes as n increases. The horizontal axis is n, the vertical axis is period, and color represents power in decibels. A horizontal band of warm color indicates a stable periodicity persisting across the data. A band that drifts, appears, or disappears indicates that the periodicity is transient or evolving.

12 Common patterns and their interpretation

Pattern 1: No periodicity (e.g., a monotone growth curve)

The power spectrum shows no peaks above the confidence thresholds. The multi-scale table is empty or reports only weak ACF values. The summary says "no statistically significant periodicity."

Pattern 2: Clean sinusoidal oscillation

A single dominant FFT peak matches the fundamental period from the multi-scale analysis. Harmonics are absent or very weak. The summary reports the period and a high ACF strength.

Pattern 3: Non-sinusoidal oscillation (e.g., sawtooth)

The FFT table shows multiple significant peaks. The highest-power FFT peak may have a short period (a high harmonic), but the multi-scale analysis identifies a longer fundamental period, and the harmonic analysis table shows that the FFT peaks are integer multiples of it. The summary explains that the dominant FFT peak is a harmonic and that the waveform is non-sinusoidal.

Pattern 4: Periodicity with drift

The detrending selects Stage B (Savitzky–Golay). After detrending, periodicity may or may not be present. The spectrogram can reveal whether the periodicity is stable or confined to part of the data range.